

PROJECT SYNOPSIS

GSM & IOT Based Smart Agriculture Monitoring System

BACHELOR OF TECHNOLOGY

Electronics and Telecommunication Engineering

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(Academic Year: 2025- 26)

CERTIFICATE

“The Project report entitled “GSM & IOT Based Smart Agriculture Monitoring System”

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Table of Content

Sr. No	Content	Page No
1.	Introduction	4
2.	Objective and Scope of the Project	6
3.	Justification	7
4.	Literature Review	8
5.	Methodology	11
6.	Software Required	14
7.	Hardware Required	15
8.	References [IEEE Format]	16

1. INTRODUCTION

Most important factors for the quality and productivity of plant growth are temperature, humidity and light. Continuous monitoring of these Environmental variables provides valuable Information to the grower to better Understand how each factor affects growth and how to maximize crop productiveness. The optimal greenhouse micro climate Adjustment can enable us to improve Productivity and to achieve remarkable Energy savings especially during the Winter in northern countries. WSN Composed of hundreds of nodes which have ability of sensing, actuation and communicating, has great advantages in terms of high accuracy, fault tolerance, flexibility, cost, autonomy and robustness compared to wired ones. Moreover, with the onset of IOT and M2M Communications, it is poised to become a very significant enabling technology in many sectors, like military, environment, health, home and other commercial areas.

IOT is a general term, covering a Number of technologies that allows devices to communicate with each other, with or without human intervention. This paper presents a novel approach to implement Wireless greenhouse automation and Monitoring system which in a timely manner provides a possibility for screen Monitoring of detailed data about the Conditions of the greenhouse. Furthermore, the suggested setup can be incorporated with other internet and messaging services (i.e. Web, WAP, SMS) to provide Communication for farmers. The wireless Sensor network (WSN) is one of the most significant technologies in the 21st century and they are very suitable for distributed data collecting and monitoring in tough environments such as greenhouses. The Other most significant technology in the 21st century is the Internet of Things (IOT) Which has rapidly developed covering Hundreds of applications in the civil, Health, military and agriculture areas.

In Modern greenhouses, several measurement Points are required to trace down the local Climate parameters in different parts of a Large-scale greenhouse in order to ensure Proper operation of the greenhouse Automation system. Cabling would make the measurement system expensive, Vulnerable and also difficult to relocate Once installed. This paper presents a WSN Prototype consisting of Micaz nodes which are used to measure greenhouses Temperature, light, pressure and humidity. Measurement data have been shared with the help of IOT. With this system farmers can control their greenhouse from their Mobile phones or computers which have Internet

ABSTRACT

“Internet of Things” (IoT) is a technology that permits things to communicate and connect with each other. This is helpful in Changing the patterns and processes in both industry and agriculture towards higher efficiency. A system is proposed which Describes the smart farming in order to improve the production process in planting. Smart farming consists of two main parts Which are a sensor system and a control system. Sensor system consists of a set of tools to obtain the sensed values. Control System involves a blower, watering and roofing system operated on a human interface. Two Arduino boards are programmed for sensing and the controlling system. Programming for controlling the system is done in python. The sensed values from the Different sensor are viewed on an LCD display as well as a serial monitor. Results are maintained as a database in excel sheet and the graphical representation of the same is obtained. Looking on to results obtained from the sensor system, the control System is activated using the python controlling console. An increase in product quality and quantity is achieved by following the proper decision-making process.

Agriculture sector in India is diminishing day by day which affects the production capacity of ecosystem. There is an exigent need to solve the problem in the domain to restore vibrancy and put it back on higher growth. The Concepts of Internet of Things (IOT) are applied to a number of applications ranging from home automation to IOT, where connecting Physical things, from anywhere critical parameters is enhanced with email and short messaging system for alert system. Whereas, Greenhouse is controlled area-controlled area environment to grow plants. The main aim of this project is to Design a simple, low cost, Arduino based system to monitor the values of environmental parameters and that are Continuously uploaded and controlled in order to achieve optimum plant growth and yield. A GSM (Global System for Mobile communication) modem is used to send SMS which displays the present status of the environmental parameter. We Also use prediction for automatically direction and resolution of any problem in the devices. Various sensor nodes are Deployed at different locations in the farm. Controlling these parameters are through any remote device or internet services and the operations are performed by interfacing sensors, Wi-Fi, camera with microcontroller. IOT is a shared Network of Objects where these objects interact through Internet. Fluctuations in rainfall or market prices can cause profits to quickly Rise or plummet. Obtaining accurate, ongoing data on operations has historically also been a challenge.

2. Objective and Scope of the Project

- The objective of a GSM & IoT based smart agriculture monitoring system is to transform traditional farming into a data-driven, efficient, and sustainable practice by enabling farmers to monitor and manage their fields remotely.
- This is achieved by collecting real-time data on critical environmental and soil parameters, such as soil moisture, temperature, and humidity, using a network of sensors.
- This data is transmitted via IoT to a cloud platform and can also trigger SMS alerts via GSM to the farmer's mobile phone for immediate attention to any abnormal conditions.
- The system aims to optimize the use of resources like water and electricity through automated irrigation, reduce manual labor, and ultimately increase crop yields and overall farm productivity.
- The scope of this project encompasses the design, development, and implementation of an integrated hardware and software solution for precision agriculture.

3. Justification

- Increased crop yield and quality: By providing real-time data, the system helps farmers optimize growing conditions, leading to better crop health and higher yields.
- Efficient resource management: Automated irrigation based on real-time soil moisture prevents both over and under-watering, conserving water and reducing costs.
- Reduced labor and time: Remote monitoring and automated alerts reduce the need for constant manual checks, saving farmers time and effort.
- Timely alerts and interventions: The GSM component sends SMS alerts when critical thresholds are met, allowing for immediate action to prevent problems like crop damage or disease.
- Informed decision-making: Data is collected, processed, and displayed on a web application, giving farmers a clear, data-driven overview of their field conditions from anywhere.
- Sustainable agriculture: The efficient use of resources and reduced waste contribute to more sustainable farming practices, which is crucial for meeting growing food demands.
- Scalability and future-proofing: The IoT foundation allows for the integration of more advanced technologies, such as machine learning and AI, in the future to further optimize farm management.

4. LITERATURE SURVEY

[1] Adithya Vadapalli, Swapna Paravel and Venkata Rao Dadi. [2020]

In this paper the author discusses existing agriculture practices. With the compilation of data from sensors, actuators and modern electronic gadgets the farmer can major agricultural fields. Smart agriculture can forecast weather data, switching ON the pump motor acknowledging the dampness of soil in terms of moisture levels with the help of sensors which are interfaced to process module Arduino UNO. The smart agriculture system can be operated anywhere with the help of networking technology studied.

[2] Muth Noori Naresh, P Munaswamy. [2019]

In this paper author proposes thought of consolidating the most recent innovation into agrarian field to turn the customary techniques for water system to current strategies in the way making simple profitable and temperature trimming. Some degree of mechanization is presented empowering the idea of observing the field and the product condition inside some long separate extends utilizing cloud administrations. The points of interest like water sparing and water sparing are started utilizing sensors that work consequently as they are modified. This idea of modernization of farming is straightforward, reasonable and operable. As relying upon these parameters is team esteemed rancher can without much of stretch choose which fungi sides and pesticides are utilized for enhancing crop creation.

[3] MD Safayet Ahmad and Akhlak Uz Zaman. [2020]

In this paper author has described it is design and implementations of their IOT based monitoring system. A device combined with sensors node and sink node, a server, and a smartphone application is used for this system. GSM module, Bluetooth module, SD card module, and various sensors to acquire data from agriculture field and transmitted to a server through GSM network. On the other end, the web interface written in PHP is implemented to connect to database directly. And a smartphone application has been developed to display all the data. Furthermore, this implementation is low cost.

[4] Mythili, Meenakshi Kumari, Aproov Tripathi, Neha Pal. [2019]

In this paper the author endorses sensible IOT based Agriculture stick as farmers aid by obtaining live knowledge (temperature, soil moisture) of farm data. These live readings help the farmers to try clever farming and to increase their average crop yields, also the quality of plants. The smart agriculture with Arduino technology supports the farmers to control the live farm data and get the desired crop cultivation results.

[5] R. Bulli Babu, CH. Jonathansoumith, T. Charisma Sri Lakshmi and R. Keshav Rao. [2015]

In this paper an author proposes a farming and government observing framework for checking data concerning an outside by using wireless sensor network (WSN) innovation. The proposed rural environment observing server framework gathers natural and soil data on the outside through (WSN) based ecological and soil sensors. In this paper using soil moisture sensor and temperature sensor. This sensor helps the field to control the water level and temperature. Here they have used a wireless sensor network as GSM (global system for mobile communication).

[6] Carolin EI Fiorenza, Sushmita Sharma, Soumya Ranjan, Shashank. [2018]

In this paper the author has used components for the supply of the required current for +5 volt. They have been used for sensors which are: DHT11 sensor, LDR sensor, Soil moisture sensor and PH sensor. A fan, exhaust fan, water pump, artificial light and motor pumps are attached to the Arduino for cooling purposes. So that the user can easily monitor and control the parameters through android mobile application. For which Android application is developed. The user can also log on the application by providing username and password for verification. The following data of the user is stored in the database. A GSM modem is also used to send text messages which immediately displays the current status of the environmental parameters. The DHT11 sensor is used to analyze the excess of temperature and humidity in the environment. Where high temperature affects the availability of crops, to produce fruits. Hence the whole system sends SMS alerts to the various mobile users.

5. PROPOSED WORK

The sensors and the devices are connected to the Arduino as shown in Fig 1. consist of ATmega328 microcontroller. Thus, voltage output from the sensors sent as inputs into the Arduino. Based on the input values, Arduino output specifies voltage to turn ON/OFF the devices the proposed system as shown in the Block diagram is serves two purposes:

- Gardening purpose
- Agriculture purpose.

The Heart of this system is the Arduino board. Whenever there is a sufficient amount of water in the field, water will not get pumped. Whenever the water moisture level of the soil goes low, water will get pumped until fulfilment of the amount of moisture is required. Soil sensor measures the soil value of the field thus the sensor values are continuously monitored and the readings are displayed to the farmer's mobile via GSM sim900A module where a sim with 3G data pack is inserted into this modem which provides IOT features to the system.

The water sensor will monitor the amount of water available in the tank. The temperature sensor will monitor the surrounding temperature which is mostly useful, when this system used in Greenhouse or Polyhouse. LCD is used to show monitored values measured by sensors and which is useful when farmer reached on field. Webpage interfaced with system to monitored all measured values from farmer's desktop or Laptop. This facility enables farmers to check and monitor all measured values from remote locations.

This system will not only provide the facility to farmers to monitor Temperature, Water moisture level, Tank water level but also it reduces work efforts required to stay physically at remote site. Due to these farmers can concentrate on other field work and this directly increases their work efficiency.

I. Objectives

- ❖ To manage & their competitive advantages compared to other bio-products.
- ❖ To enable Monitoring and Protection System Using IOT.
- ❖ To implement a system for farm Health Monitoring.
- ❖ To implement Water Waste Management by Sensor Network.
- ❖ To monitor & Control Soil Moisture for Crop Health.
- ❖ To build a smart system for farmers to increase their work efficiency. To build an Environment friendly & Supportive smart irrigation system.

II. Methodology

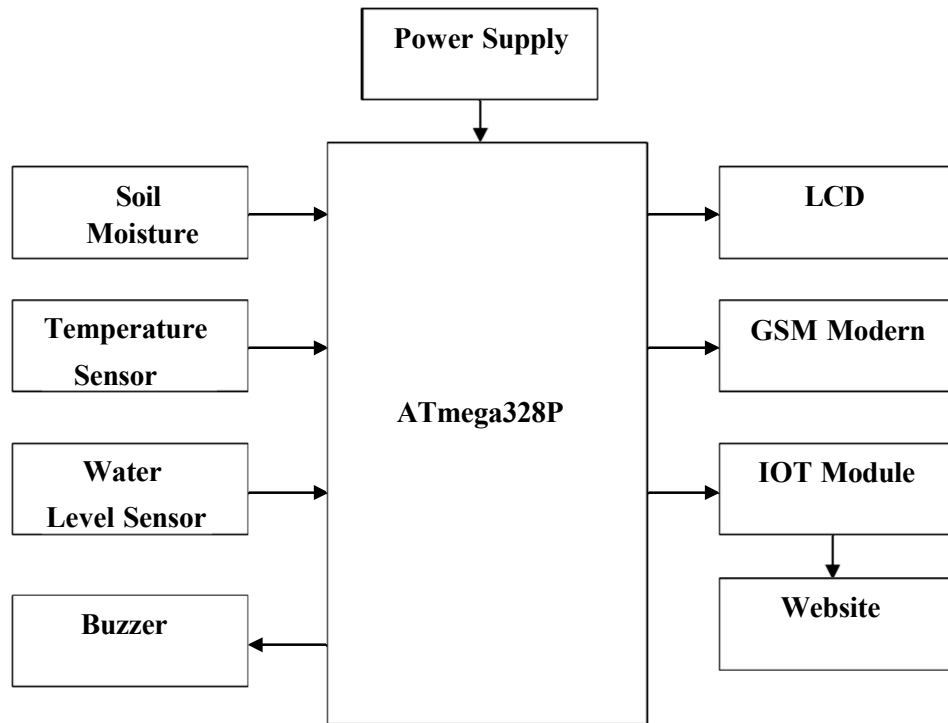


Fig 1. Block Diagram of Proposed System

The sensors and the devices are connected to the Arduino as shown in the Fig.1 consist of ATmega328 microcontroller. Thus, voltage output from the sensors sent as inputs into the Arduino. Based on the input values, Arduino output specifies voltage to turn ON/OFF the devices the proposed system as shown in the Block diagram serves two purposes: one, for gardening purpose and another, for agriculture purpose. The heart of the project is the Arduino Uno board and shown in the block diagram. Whenever there is a sufficient amount of water in the field, water will not pump into the field. Whenever the water moisture level of the soil goes low, water will be pumped into the field until the required amount of moisture is required. Soil sensor measures the soil value of the field thus the sensor values are continuously monitored and the readings are displayed to the farmer's mobile via GSM sim900A module where a sim with 3G data pack is inserted into this modem which provides IOT features to the system.

III. IOT TECHNOLOGY AND AGRICULTURE

1) Arduino

The Arduino UNO is a widely used open-source microcontroller board based on the ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board features 14 Digital pins and 6 Analog pins.



Fig 2. Arduino UNO

2) Soil Moisture Sensor

The Moisture sensor is used to measure the water content (moisture) of soil. When the soil is having water shortage, the module output is at high level else the output is at low level. This sensor reminds the user to water their plants and also monitors the moisture content of soil. It has been widely used in agriculture, land irrigation and botanical gardening.

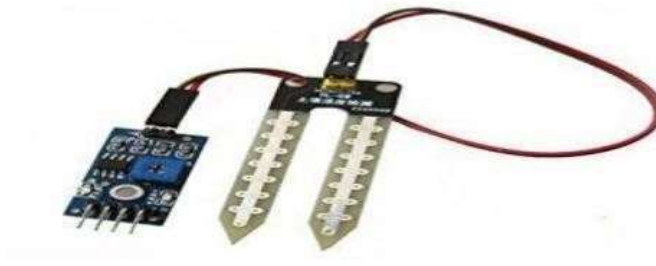


Fig 3. Soil Moisture Sensor

3) Temperature Sensor LM35

LM35 is a temperature measuring device having an analog output voltage proportional to the temperature. It provides output voltage in Centigrade (Celsius). It does not require any external calibration circuitry. The sensitivity of LM35 is 10 mV/degree Celsius. As temperature increases, output voltage also increases.



Fig 4. Temperature Sensor LM35

4) Water Level Sensor

The working of the water level sensor is pretty straight forward. The series of exposed parallel conductors, together acts as a variable resistor (just like a potentiometer) whose resistance varies according to the water level. The change in resistance corresponds to the distance from the top of the sensor to the surface of the water. The resistance is inversely proportional to the height of the water: The more water the sensor is immersed in, results in better conductivity and will result in a lower resistance. The less water the sensor is immersed in, results in poor conductivity and will result in a higher resistance. The sensor produces an output voltage according to the resistance, which by measuring we can determine the water level.

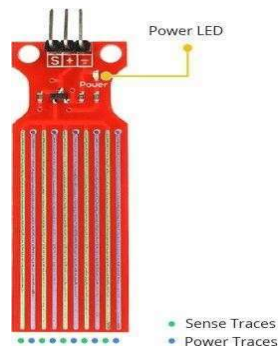
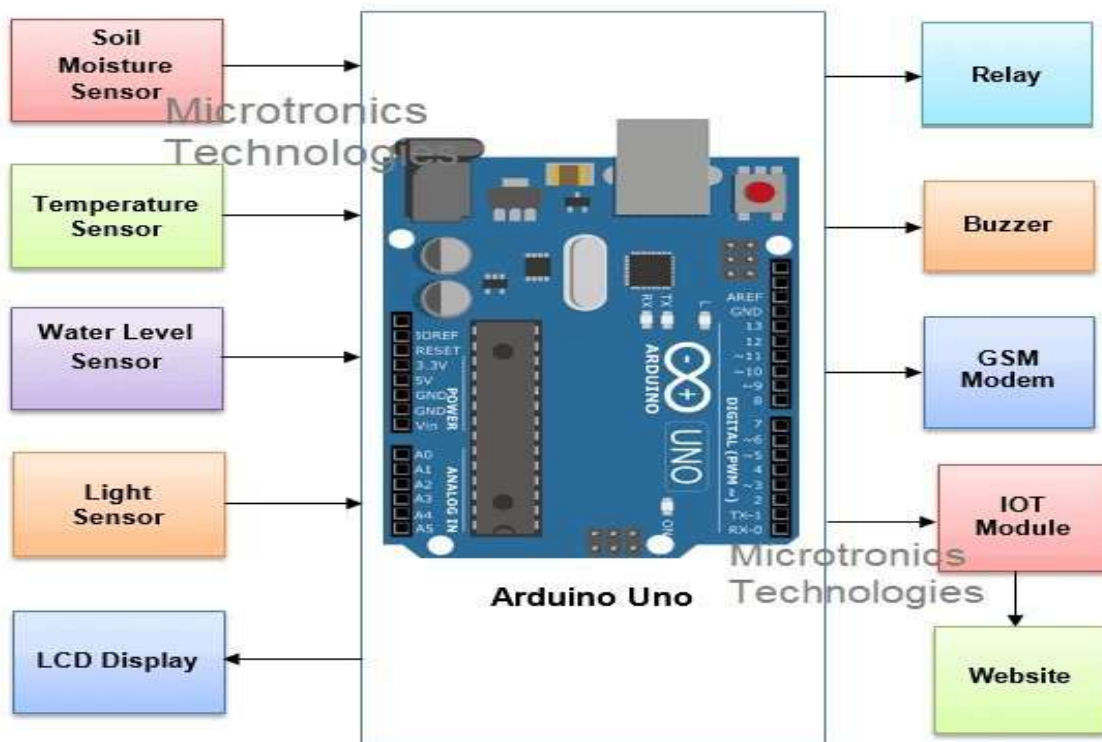


Fig 5. Water Level Sensor

6. Software Requirements

1. Arduino Uno
2. Web Application



7. Hardware Requirements

- 1. Microcontroller**
- 2. Sensors**
- 3. Communication Module: GSM modem**
- 4. Actuators**
- 5. Buzzer**
- 6. LCD Display**

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